

Ayush Mishra

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LinkedIn | GitHub | LeetCode | Portfolio

Summary

AI/ML Engineer with expertise in deep learning, NLP, and Computer Vision. Specialized in transformer architectures (BERT, ALBERT), multimodal pipelines, and Retrieval-Augmented Generation (RAG) systems. Experienced in model fine-tuning, transfer learning, and deploying AI applications using PyTorch, Hugging Face, and the MERN stack. GATE CSE 2026 qualified.

Education

United Institute of Technology <i>Bachelor of Technology (B.Tech) in Computer Science and Engineering – CGPA: 8.31/10.0</i> New R.S.J Public School <i>Class 12th (CBSE): 83.2% Class 10th (CBSE): 89.4%</i>	Expected May 2027 Prayagraj, India 2020 – 2022 Prayagraj, India
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Skills

AI/ML: PyTorch, Hugging Face Transformers, BERT, ALBERT, Fine-Tuning, Transfer Learning, Vector Embeddings
NLP & CV: Cognitive Distortion Detection, Facial Emotion Recognition, DeepFace, OpenCV, Sentence-Transformers
Generative AI: RAG, ChromaDB, LLM Integration, Semantic Search, Vector Databases
Languages: Python, Java, C, JavaScript, SQL
Web Stack: React.js, Node.js, Express.js, FastAPI, MongoDB, Tailwind CSS, Three.js, REST APIs
Core CS: DSA, OOP, Operating Systems, Git, GitHub, NumPy, Pandas

Experience

Collab4Good <i>Global Innovation Intern</i> - Developed a full-stack MERN application for NGO resource sharing, improving cross-team workflow efficiency across 5 organizations serving 200+ end users. - Engineered responsive React components and RESTful APIs within an international agile team, delivering milestones 2 sprints ahead of schedule and reducing page load times by 40%.	Aug 2025 – Present Remote / Bangkok, Thailand
United Institute of Technology <i>Software Engineering Summer Trainee</i> Built a browser-based text editor with session state persistence using localStorage, eliminating data loss across page reloads for 100+ daily active users.	Jul 2024 Prayagraj, India

Projects

NeuroWell – Multimodal Emotional Support System <i>Python, PyTorch, BERT, RAG, ChromaDB, OpenCV</i> - Architected an end-to-end multimodal AI pipeline integrating speech transcripts, facial expression analysis, and text embeddings for real-time emotional support, processing 3 input modalities simultaneously. - Engineered a RAG system with ChromaDB vector embeddings and cosine similarity retrieval to deliver rubric-grounded contextual responses, improving relevance by 35% over baseline.	2025
Legal Lens – AI Legal Document Analyzer <i>React, FastAPI, Sentence-Transformers, ChromaDB, Three.js</i> - Engineered an asynchronous AI agent system scanning legal contracts for high-risk GDPR/HIPAA clause violations in real time using NLP-based text classification and semantic search. - Built a compliance engine with ChromaDB vector database seeded with regulatory standards and a custom scoring model combining cosine similarity with keyword-penalty weighting. - Implemented WebSocket live updates with automatic HTTP-polling fallback and an interactive 3D vector visualizer using Three.js for embedding space exploration.	2025
AI Interview Coach – Multimodal Interview Platform <i>React, Node.js, FastAPI, OpenCV, BERT, MongoDB</i> - Designed a 3-tier microservice architecture (React frontend, Node.js API, Python FastAPI ML backend) delivering real-time multimodal interview evaluation. - Integrated OpenCV facial expression tracking at 1Hz and a fine-tuned BERT model for filler-word detection, achieving 92% classification accuracy on validation data. - Developed a timestamp-synchronized signal fusion algorithm computing weighted composite confidence scores and an interactive Three.js scrubbing timeline for session review.	2025

Honors and Achievements

- GATE CSE 2026 Qualified** – Cleared Graduate Aptitude Test in Engineering for Computer Science.
- Global Merit Scholar** – Selected for AIT Bangkok with 90% merit scholarship among international applicants.
- Competitive Programming** – 150+ day LeetCode streak solving advanced DSA problems; active on Codeforces.
- Leadership** – Event Head for Enigma College Fest (5,000+ attendees); Core Organizer for UHack Hackathon.